

האוניברסיטה העברית בירושלים
THE HEBREW UNIVERSITY OF JERUSALEM

On Comparing Internal Representations in Brains and Machines

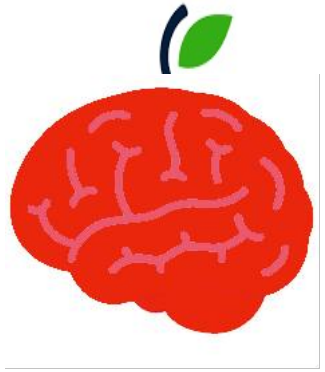
Daniella Har-Shalom & Yair Weiss

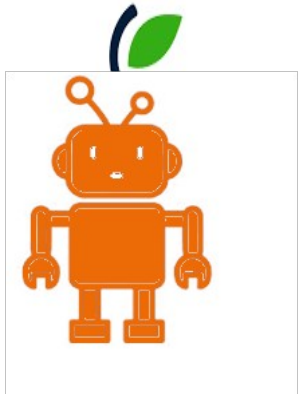
Gatsby Meeting

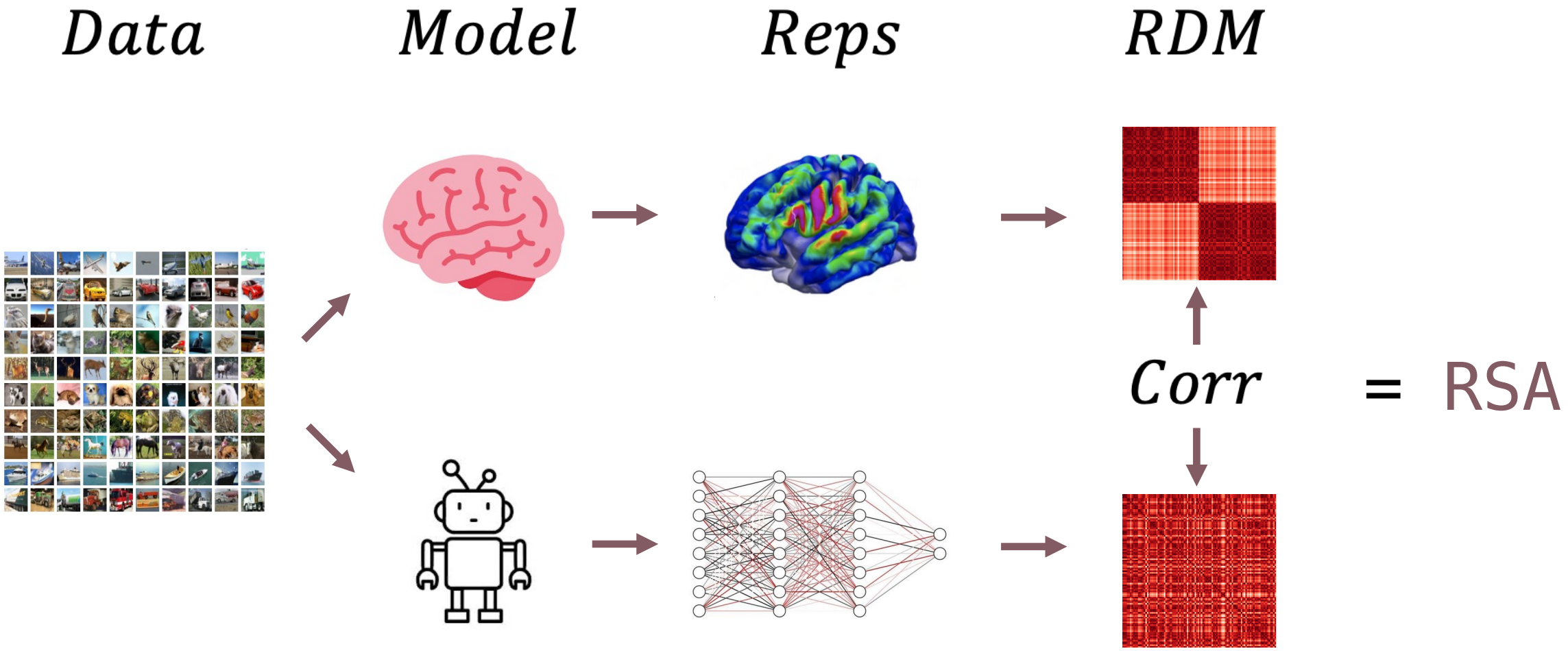
June 2026

Motivation

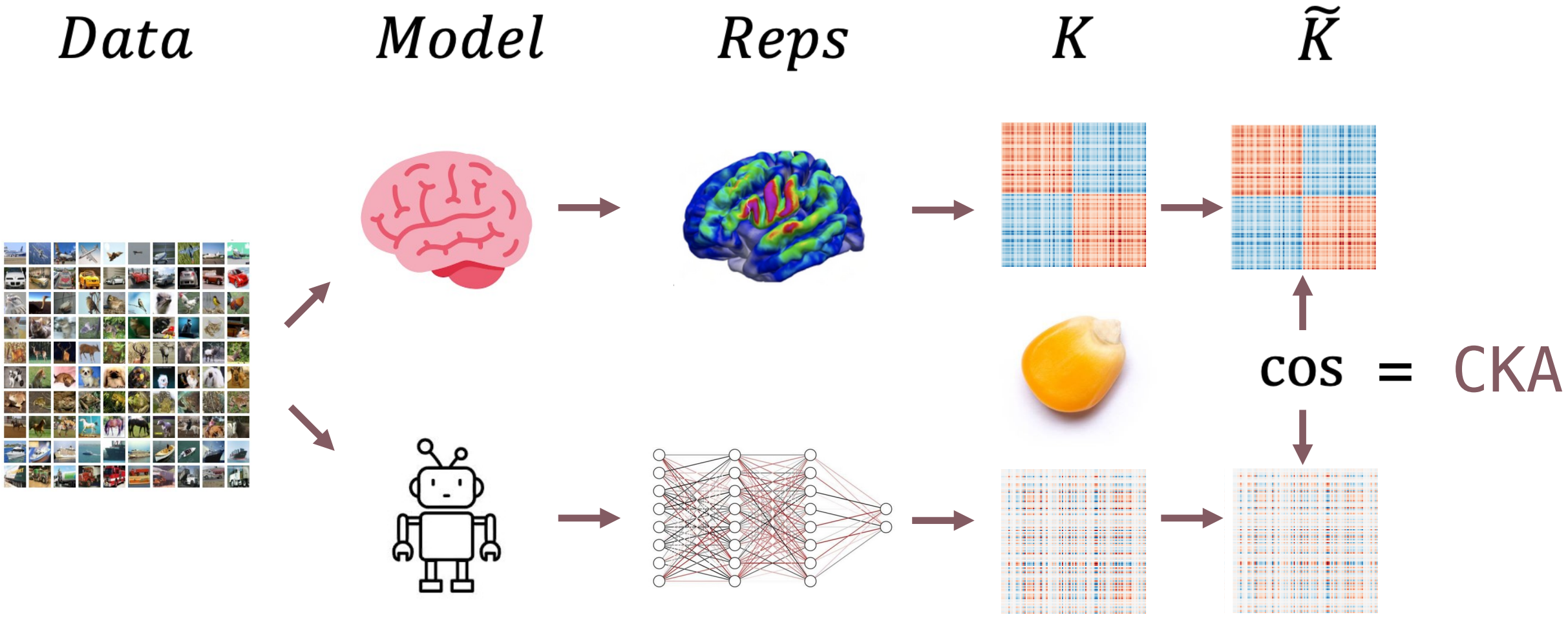
- **Comparing different systems**
  - How do we know if two systems represent the world similarly?
- **Standard approach:**
  - Present stimuli > Extract representations > Compare with a metric
- Not trivial
- Choice of stimuli and metric defines what "similar" means



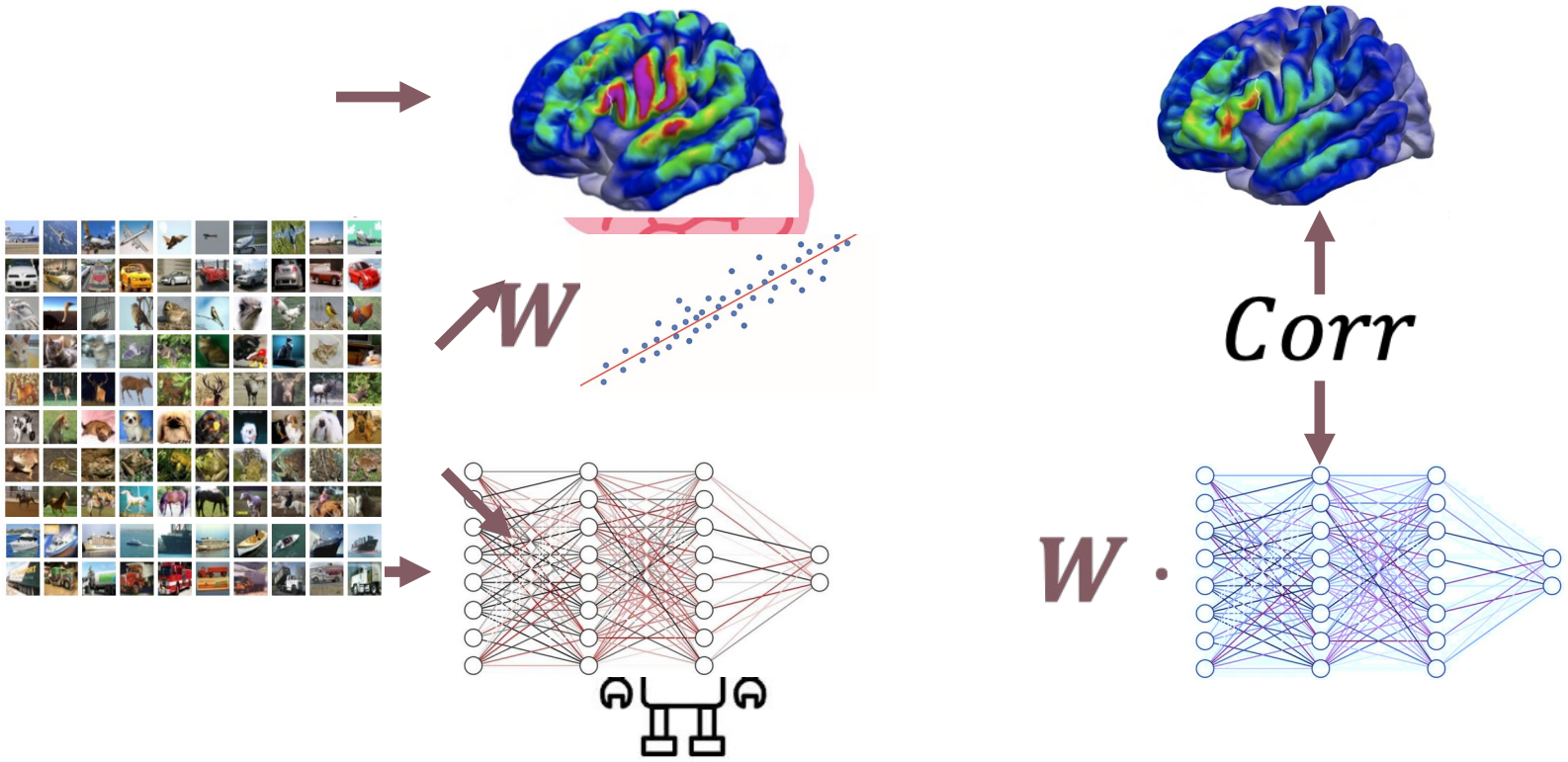




Kriegeskorte et al. 2008



Kornblith et al. 2019; Cortes et al. 2012; Cristianini et al. 2002

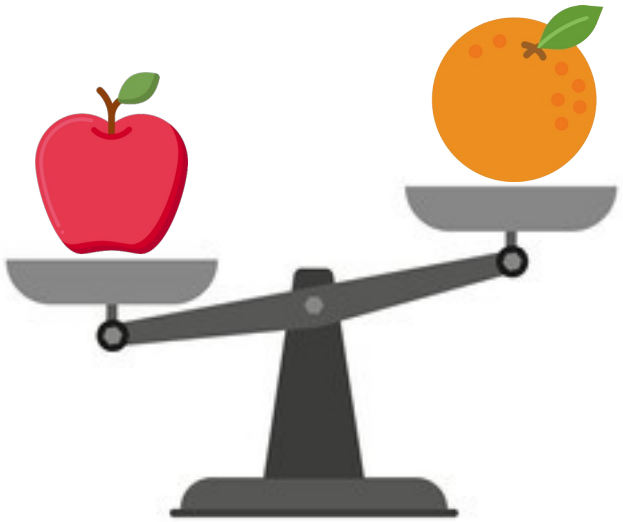


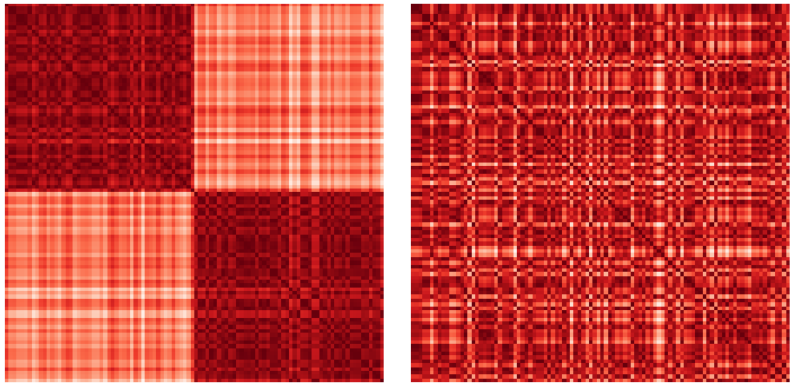
Brain Score = (Neural Predictivity)

Schrimpf et al. 2018; Yamins et al. 2014

Why these methods make sense

- Bypass the dimensionality problem
- Identical systems → $score = 1$
- Invariant to orthogonal transformations
- Interpretable(ish)





Widely used

Representational similarity analysis-connecting th
neuroscience
N **Kriegeskorte**, M Mur, PA Bandettini - Frontiers in systems …,
A fundamental challenge for systems neuroscience is to quantita
branches of research: brain-activity measurement, behavioral me
computational modeling. Using measured brain-activity patterns to evaluate computational
network models is complicated by the need to define t
of the model and the channels of the brain-activity dat
from functional magnetic resonance imaging (fMRI). S
☆ Save 99 Cite Cited by 5085 Related articles

[PDF] Algorithms for learning kernels based on centered alignment
C **Cortes**, M Mohri, A Rostamizadeh - The Journal of Machine Learning …, **2012** - jmlr.org
This paper presents new and effective algorithms for learning kernels. In particular, as shown by our empirical results, these algorithms consistently outperform the so-called uniform …
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Brain-score: Which artificial neural network for object recognition is most brain-like?
M Schrimpf, J Kubilius, H Hong, NJ Majaj… - BioRxiv, 2018 - biorxiv.org
The internal representations of early deep artificial neural networks (ANNs) were found to be remarkably similar to the internal neural representations measured experimentally in the …
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Similarity of neural network representations revisited
S **Kornblith**, M Norouzi, H Lee… - … conference on
Recent work has sought to understand the behavior o
representations between layers and between different
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Performance-optimized hierarchical models predict neural responses in **higher visual cortex**
DLK Yamins, H Hong, CF Cadieu, EA Solomon… - Proceedings of the …, 2014 - pnas.org
… **models** of **neural responses** in **higher** ventral **cortex**. Our approach operationalizes a hypothesis for how two biological constraints together shaped **visual cortex**… the **hierarchical** network …
☆ Save 99 Cite Cited by 2849 Related articles All 23 versions

A closer look

A large-scale examination of inductive biases shaping high-level visual representation in brains and machines (Conwell et al. 2024, Nature Communications)

Goal: reveal relationships between model variations and brain-predictive representation

- “transformers were on average less **predictive** than the CNNs”
- “models trained on ‘larger, more diverse datasets’ were the best **predictors**”
- “raw quantity of training images does not necessarily lead to increased **brain-predictivity**”
- “ImageNet diet leads to the highest overall **brain predictivity**”
- “randomly initialized models may, in some cases, be as **predictive of the brain** as trained models”

“the difference between the metrics constitutes one of the most substantial effects on brain predictivity of any we assess”

Prior work

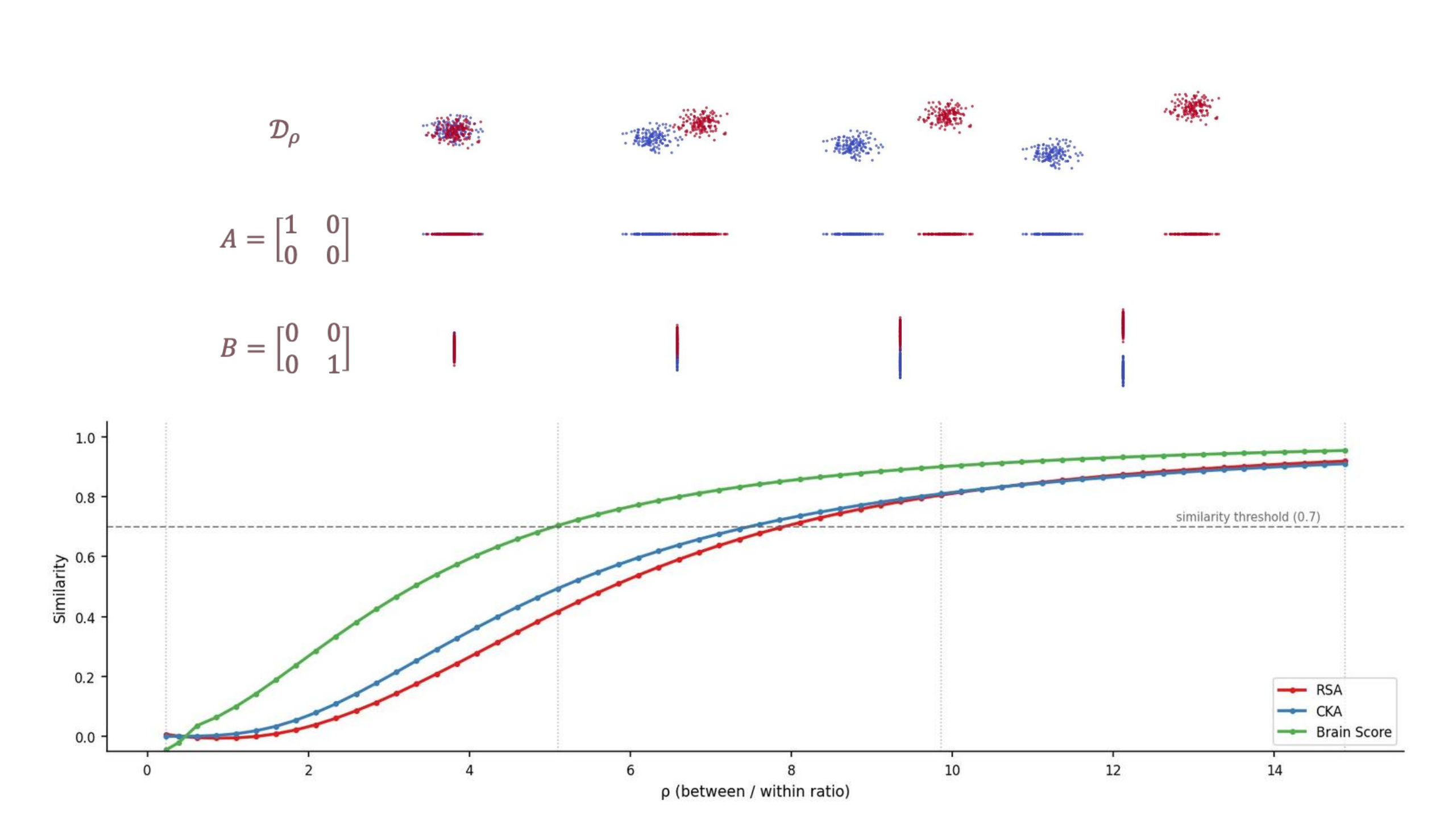
- Scores fail to distinguish models / depend on the data
  - Dujmović et al. 2022, Cloos et al. 2024, Kuoch et al. 2024
- Brain-Score is too permissive
  - Feghhi et al. 2024, Soni et al. 2024, Conwell et al. 2024
- These metrics are dominated by top variance directions
  - Kornblith et al. 2019, Nguyen et al. 2021, Ding et al. 2021, Davari et al. 2022, Cloos et al. 2024

Toy experimen



- **Setup**
  - Two isotropic Gaussians $\mathcal{N}\left(\pm\frac{\delta}{2}\mathbf{u}, \sigma^2 I\right)$ in $\mathbb{R}^2$
  - $\rho = \frac{\delta}{\sigma}$ (“*separation ratio*”)
  - Two linear models A, B

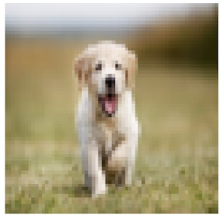
$$A = \begin{bmatrix} 1 & 0 \\ 0 & 0 \end{bmatrix}, \qquad B = \begin{bmatrix} 0 & 0 \\ 0 & 1 \end{bmatrix}$$


$\mathcal{D}_\rho$
$A = \begin{bmatrix} 1 & 0 \\ 0 & 0 \end{bmatrix}$
$B = \begin{bmatrix} 0 & 0 \\ 0 & 1 \end{bmatrix}$
Similarity
ρ (between / within ratio)
similarity threshold (0.7)
RSA
CKA
Brain Score
0.0
0.2
0.4
0.6
0.8
1.0
0
2
4
6
8
10
12
14

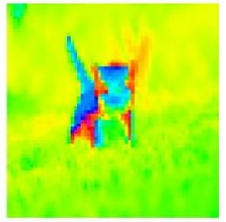
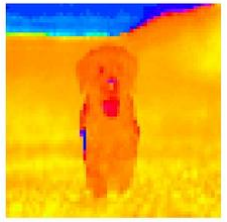
Slightly less toyis 

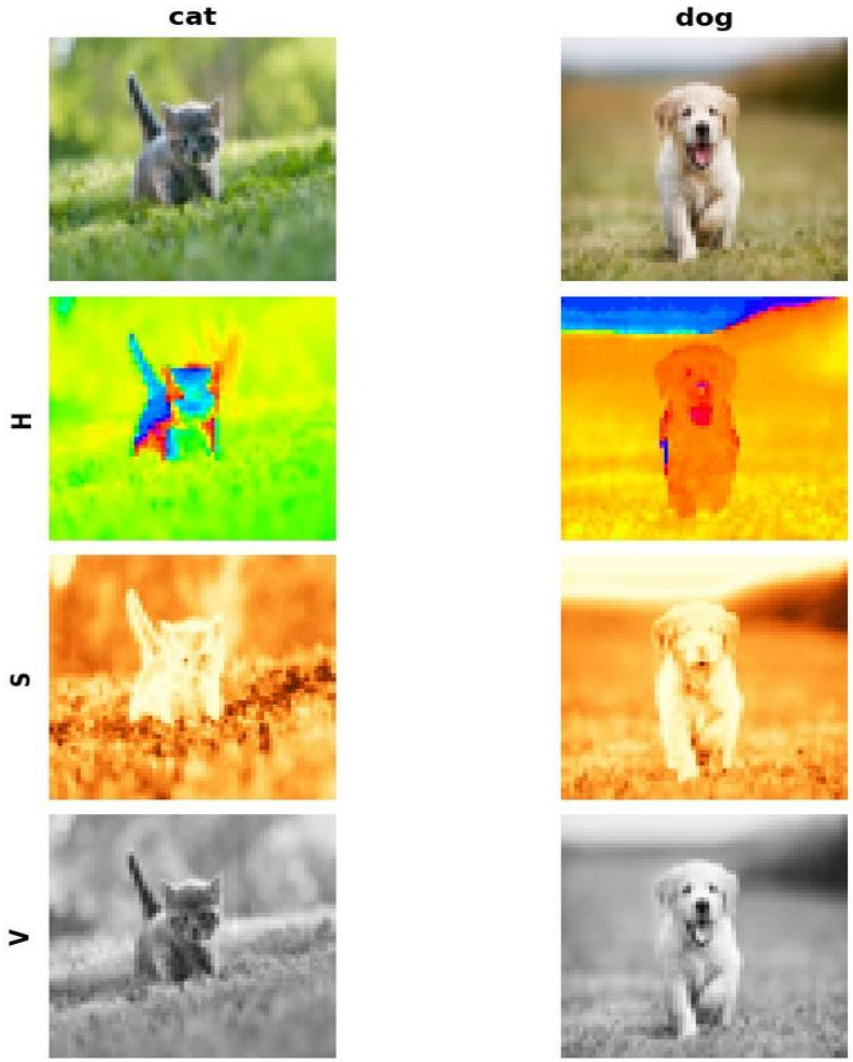
- **Setup**
  - Two isotropic "Image" Gaussians in $\mathbb{R}^d$

$$\frac{1}{2}\mathcal{N}(\boldsymbol{\mu_i}, \sigma^2 I) + \frac{1}{2}\mathcal{N}(\boldsymbol{\mu_j}, \sigma^2 I)$$

$\boldsymbol{\mu_i} =$  $\boldsymbol{\mu_j} =$ 

  - Separation ratio $\rho = \frac{\delta}{\sigma}$, where $\delta = \| \boldsymbol{\mu}_i - \boldsymbol{\mu}_j \|$
  - Three models H, S, V - each projects onto one HSV channel:

$H =$   $V =$   $S =$  


cat
dog
H
S
V

A controlled case: two Gaussians

- **Setup:** Two isotropic Gaussians $\mathcal{N}\left(\pm\frac{\delta}{2}\mathbf{u}, \sigma^2 I\right)$ in $\mathbb{R}^d$, separation ratio $\rho = \delta/\sigma$, two linear models A, B
- **Theorem:** For any A, B with $A\mathbf{u}, B\mathbf{u} \neq 0$, then as $\rho \to \infty$

$$\begin{cases} RSA(A, B, \mathcal{D}_\rho) \to 1 \\ CKA(A, B, \mathcal{D}_\rho) \to 1 \\ BS(A, B, \mathcal{D}_\rho) \to 1 \end{cases}$$

- **Intuition:** As $\rho \to \infty$, both distance matrices become dominated by the cluster-indicator structure, scaled by each model's coupling to the separation direction

Real data



- **Setup**
  - Two CIFAR-10 catgories:

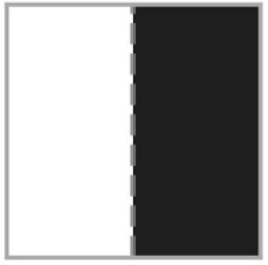
Airplanes



Ships

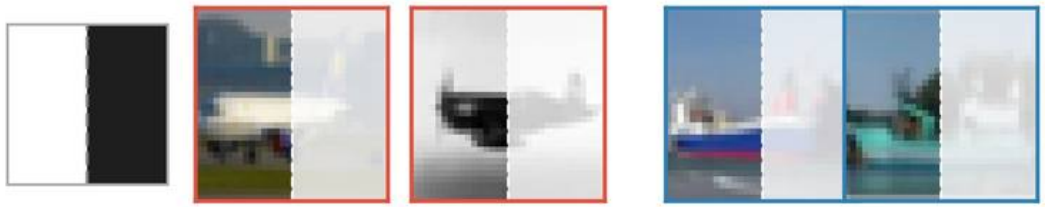


  - Two masking models:

$A =$ 

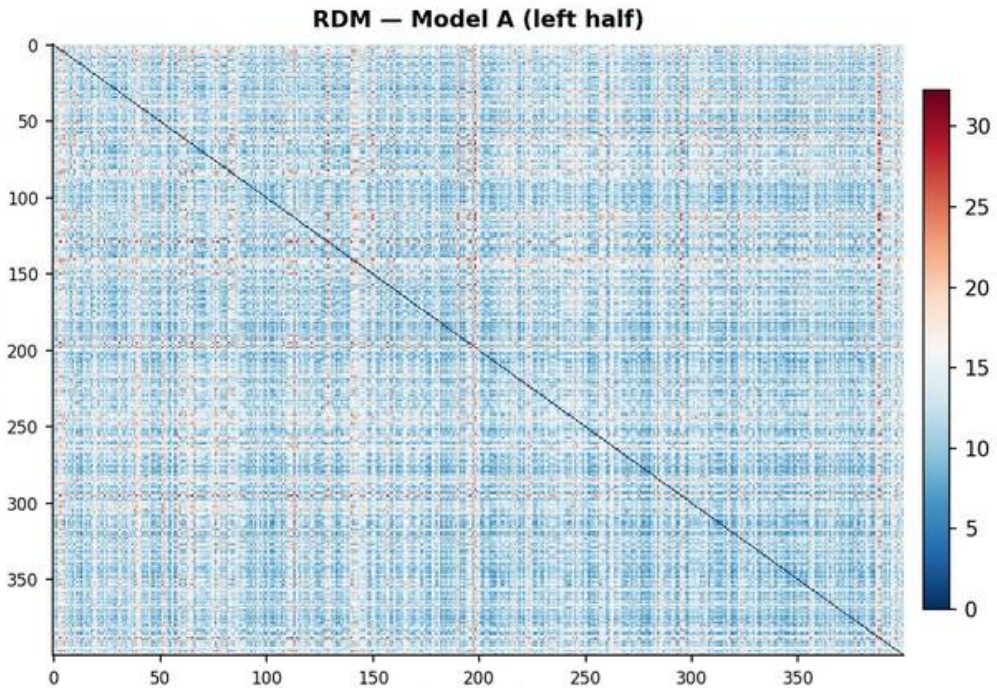
$B =$ 

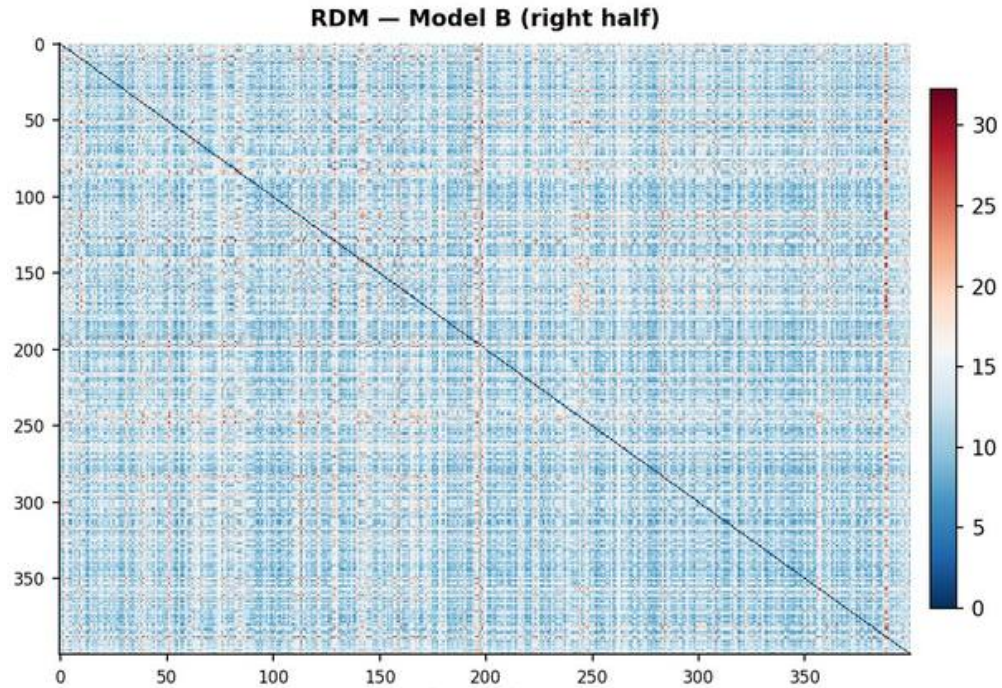
Model A



Model B







$(RSA = 0.82)$

Recap

- So far: Highly sensitive to choice of data
  - Choosing data ***carelessly*** may manipulate the score
    - Very different models can get a very high score
    - Same models can score anywhere from 0 to 1, depending on data
- On the flip side:
  - Could choosing data ***carefully*** make the score more diagnostic?
  - *It's not about manipulating the score - but matching data to the question*

Highly sensitive to choice of data

Goal: Informative data

- The score depends on which direction the data separates along
  - Separates where models **agree** → **high** score
  - Separates where models **disagree** → **low** score

- Failure modes
  - **False similarity**: high score, didn't probe where they differ
  - **False difference**: low score, direction that doesn't matter

- Probing where models could differ?

Controversial Stimuli

- **Controversial stimuli[3,4]**
  - Pair of models: maximize disagreement
  - Output / representation level
- **Adversarial examples[1] & metamers[2]**
  - Single model (vs. human perception)
  - Output level

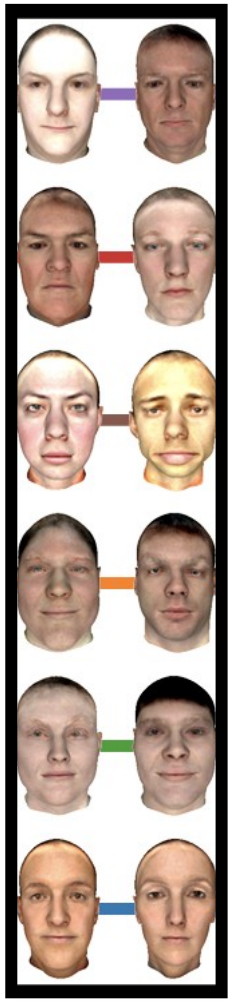
Adversarial Examples



Metamers

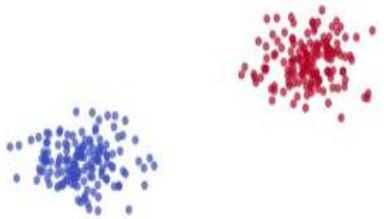


Controversial Stimuli



[1]Goodfellow et al. 2014; [2]Feather et al. 2023; [3]Golan et al. 2020; [4]Golan, Guo et al. 2022

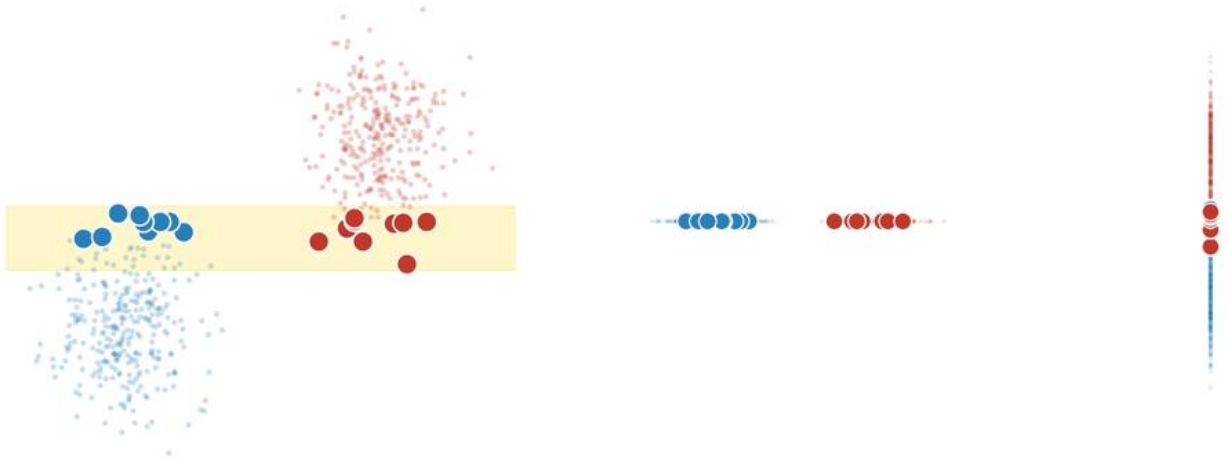
Toy data revisited

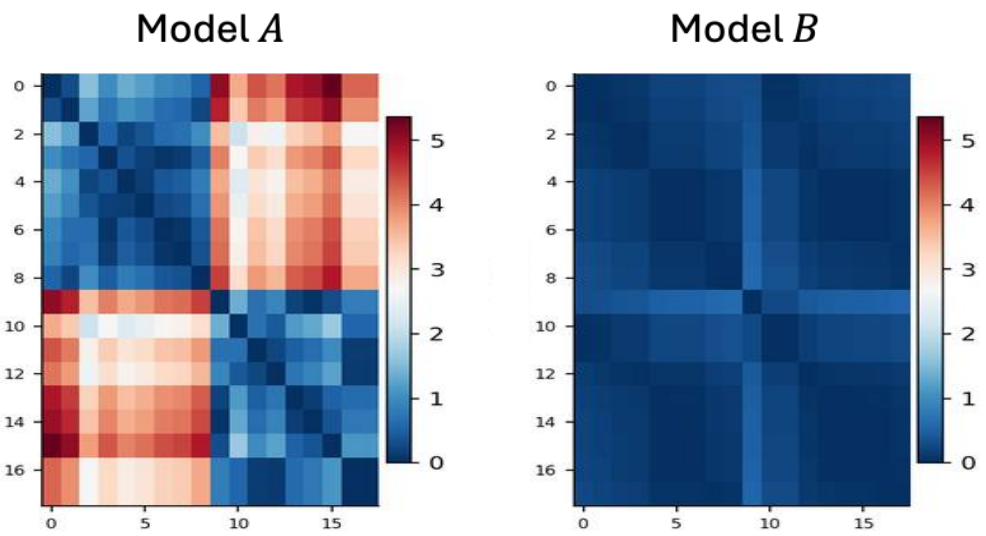


$$A = \begin{bmatrix} 1 & 0 \\ 0 & 0 \end{bmatrix}, \qquad B = \begin{bmatrix} 0 & 0 \\ 0 & 1 \end{bmatrix}$$

$\mathcal{D}_\rho$ $\qquad A = \begin{bmatrix} 1 & 0 \\ 0 & 0 \end{bmatrix}$ $\qquad B = \begin{bmatrix} 0 & 0 \\ 0 & 1 \end{bmatrix}$

RDMs

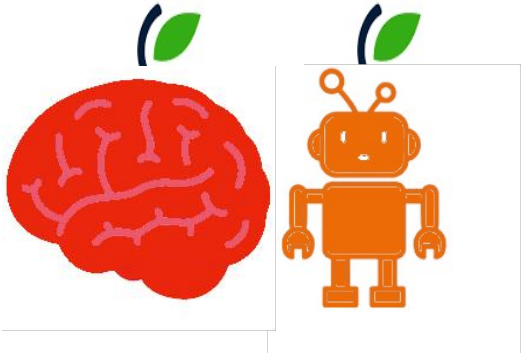




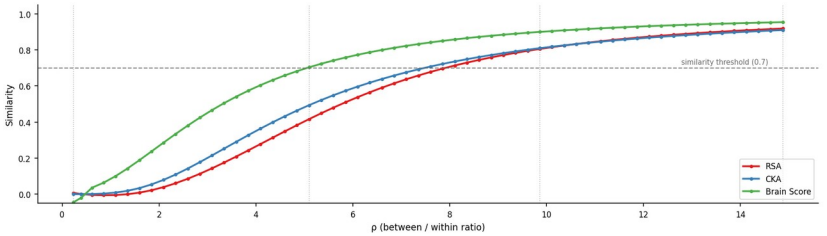
$$\Rightarrow RSA = 0.02$$

Summary

- Similarity-based methods
  - Compare representations across brains and models
  - Bypass the dimension-mismatch problem



- Highly dependent on data
  - Very different models can get a very high score
  - Same models can score anywhere from 0 to 1, depending on data



- False similarity:
  - High score, didn't probe where they differ
  - Maximize controversiality
- False difference?